

FPGA TRNG - NIST Statistical Test Suite Report

Date: 2026-08-04 20:58:05 | Port: COM8 | Capture Time: 25 sec | SHA-256 whitening: ON

1. Raw Capture Data & Bias Proof (Monobit Test)

Total Bits Captured: 765,952
Ones (1s): 383,554 (50.08%) Zeros (0s): 382,398 (49.92%)
S-value: 1.320861 P-value: 0.186548 Verdict: PASS

2. Remaining NIST SP 800-22 Tests

Test Name	P-Value	Verdict	Simple Explanation
Frequency Within Block	0.748452	PASS	Checks 50/50 balance in small blocks, not just overall.
Runs	0.068860	PASS	Checks for too many repeats of the same bit in a row.
Longest Run Ones In A Block	0.067924	PASS	Checks the longest same-bit streak isn't unusually long.
Binary Matrix Rank	0.550250	PASS	Checks a matrix built from the bits isn't oddly structured.
Discrete Fourier Transform	0.000000	FAIL	Looks for a repeating wave. Ring oscillators are one.
Non Overlapping Template Matching	0.106074	PASS	Looks for one exact short pattern recurring too often.
Maurers Universal	0.019684	PASS	Tries to compress the data. True randomness can't compress.
Serial	0.000000	FAIL	Checks if nearby bits let you guess the next one.
Approximate Entropy	0.000000	FAIL	Compares short-pattern counts to what randomness predicts.
Cumulative Sums	1.000000	PASS	Checks a running total doesn't wander too far from zero.
Random Excursion	0.683143	FAIL	Checks behaviour of a random walk built from the bits.
Random Excursion Variant	0.170252	PASS	Checks behaviour of a random walk built from the bits.

3. Overall Engineering Conclusion

Passed 9 of 13 NIST SP 800-22 tests (25-second capture, 765,952 bits). The Monobit test confirms the Von Neumann corrector removed the hardware's thermal bias (S=1.3209, P=0.1865). Remaining failures are expected: a ring oscillator is a periodic circuit, and the Von Neumann corrector fixes bit balance only, not bit-to-bit correlation. Tests that hunt for periodicity and pattern structure (e.g. spectral and template tests) detect this physical property directly. A SHA-256 whitening pass was applied to the captured stream before testing. Whitening redistributes existing entropy; it does not add entropy the source did not produce, which is why the pass count did not change substantially from the un-hashed run at the same capture duration.